

Fourier Towers, Cocycle Obstructions, and Higher Nilpotent Phase Geometry:

From Splitting Failure to $F + \Phi + H$ Data Decompositions

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Abstract

We develop a Fourier Tower interpretation of cocycle obstructions, non-splitting extensions, and higher nilpotent phase geometry.

The central idea is that the failure of a group extension to split may be reorganized, after Fourier transformation of suitable abelian layers, into phase interactions between local Fourier sectors.

Starting from an extension

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1,$$

a transversal section generally fails to preserve multiplication. This failure is measured by a cocycle. When the lower layer is Fourier transformed, the cocycle becomes a phase factor acting on characters.

Thus the guiding principle is:

splitting obstruction $\longrightarrow$ cocycle correction $\longrightarrow$ Fourier phase interaction.

We develop this viewpoint for the nilpotent group

$$UT_4(\mathbb{Z}_5),$$

whose multiplication law contains several nested cocycle corrections. These corrections generate a hierarchy of Fourier phase couplings, making $UT_4(\mathbb{Z}_5)$ a natural example beyond the ordinary Heisenberg case.

The article also discusses the distinction between internal splitting obstructions and Clifford-type splitting problems, the periodic structure of cocycle-corrected Fourier Tower operators, and possible connections with quantum circuit decompositions.

Finally, we propose an $F + \Phi + H$ decomposition principle, where F denotes local Fourier layers, Φ denotes cocycle phase interactions, and H denotes higher nilpotent propagation.

The work is primarily conceptual and organizational: its goal is not to provide a complete representation-theoretic classification, but rather to propose a recursive Fourier interpretation of cocycle interactions in solvable and nilpotent structures.

1 Introduction

Many nonabelian groups arise as extensions

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1.$$

If the extension splits, then

$$G \cong N \rtimes Q,$$

and the group structure is determined by an action of Q on N .

Semidirect products therefore provide a classical mechanism for organizing nonabelian structure.

However, in many solvable and nilpotent situations, a simple global semidirect decomposition is not available, or a chosen transversal section fails to preserve multiplication.

This failure is measured by cocycles.

The purpose of the present article is to reinterpret this phenomenon through Fourier analysis.

The central idea is:

$$\boxed{\text{failure of splitting} \longrightarrow \text{cocycle correction} \longrightarrow \text{Fourier phase interaction.}}$$

From this viewpoint, cocycles are not merely algebraic correction terms. They become phase mechanisms coupling local Fourier layers.

This leads naturally to the Fourier Tower perspective.

Instead of diagonalizing a nonabelian structure globally, one decomposes the group layer by layer, Fourier transforms the abelian quotient coordinates, and records the obstruction to splitting as phase interactions between Fourier character sectors.

The finite nilpotent group

$$UT_4(\mathbb{Z}_5)$$

provides a natural example because its multiplication law contains several nested cocycle corrections:

$$ad', \quad df', \quad ae' + bf'.$$

These terms are not accidental. They describe how lower and higher nilpotent layers interact. After Fourier transformation, they become phase factors:

$$\omega_5^{\beta ad'}, \quad \omega_5^{\epsilon df'}, \quad \omega_5^{\gamma(ae' + bf')}.$$

Thus $UT_4(\mathbb{Z}_5)$ exhibits a higher nilpotent version of Fourier phase geometry.

The article is organized as follows.

First, we recall splitting and cocycle obstructions for group extensions.

Next, we explain how cocycles become phase factors after Fourier transformation.

Then we develop the example of $UT_4(\mathbb{Z}_5)$, including an explicit computation showing the failure of a naive layerwise splitting.

We then discuss the periodic structure of Fourier Tower operators, the distinction between internal splitting and Clifford-type splitting, and possible quantum circuit interpretations.

Finally, we propose a decomposition principle of the form

$$UT_4 \rightsquigarrow F + \Phi + H,$$

where:

$$F = \text{local Fourier layers,}$$

$$\Phi = \text{cocycle phase interactions,}$$

and

$$H = \text{higher nilpotent propagation.}$$

The present work is primarily conceptual and organizational.

The goal is not to provide a complete representation-theoretic classification, but rather to propose a recursive Fourier interpretation of cocycle interactions in solvable and nilpotent groups.

2 Splitting and Cocycle Obstructions

Consider an exact sequence

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1.$$

A splitting is a homomorphism

$$s : Q \rightarrow G$$

such that the composition

$$Q \xrightarrow{s} G \rightarrow Q$$

is the identity.

In that case,

$$G \cong N \rtimes Q.$$

Thus every element $g \in G$ may be written uniquely as

$$g = n s(q), \quad n \in N, \quad q \in Q.$$

The multiplication law becomes

$$(n, q)(n', q') = (n(q \cdot n'), qq').$$

Therefore the nonabelian structure is encoded by an action of Q on N .

2.1 Failure of Multiplicativity

In general, a transversal section

$$s : Q \rightarrow G$$

need not be a homomorphism.

Instead, one typically has

$$s(q)s(q') = \omega(q, q')s(qq'),$$

where

$$\omega(q, q') \in N.$$

The map

$$\omega : Q \times Q \rightarrow N$$

is the cocycle correction measuring the defect of multiplicativity.

Thus:

$$\text{section} + \text{failure of multiplicativity} = \text{cocycle}.$$

The extension fails to behave like a global semidirect product precisely because of these correction terms.

3 Fourier Interpretation of Cocycle Corrections

Suppose now that N is abelian.

Then one may Fourier transform functions on N .

Let

$$\chi \in \widehat{N}$$

be a character of N .

Translations in N become phase multiplications after Fourier transformation:

$$n \mapsto n + n_0$$

corresponds to

$$\chi(n_0).$$

Therefore cocycle corrections become phase factors.

If

$$s(q)s(q') = \omega(q, q')s(qq'),$$

then after Fourier transformation of the N -layer, the cocycle becomes

$$\chi(\omega(q, q')).$$

Thus the extension law produces a Fourier phase interaction:

$$\boxed{\omega(q, q') \longrightarrow \chi(\omega(q, q'))}.$$

From the Fourier Tower viewpoint, nonabelianity is therefore encoded by phase couplings between local Fourier sectors.

4 Example: $UT_4(\mathbb{Z}_5)$

Consider the nilpotent group

$$UT_4(\mathbb{Z}_5),$$

consisting of upper unitriangular matrices

$$g(a, b, c, d, e, f) = \begin{pmatrix} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its multiplication law is

$$\begin{aligned} & (a, b, c, d, e, f)(a', b', c', d', e', f') \\ &= (a + a', b + b' + ad', c + c' + ae' + bf', d + d', e + e' + df', f + f'). \end{aligned}$$

The correction terms

$$ad', \quad df', \quad ae' + bf'$$

measure the failure of naive layerwise splitting.

4.1 Explicit Splitting Failure

Consider the lower central extension

$$1 \rightarrow \Gamma_2 \rightarrow UT_4(\mathbb{Z}_5) \rightarrow \Gamma_1/\Gamma_2 \rightarrow 1.$$

The quotient

$$\Gamma_1/\Gamma_2 \cong \mathbb{Z}_5^3$$

is represented by coordinates

$$(a, d, f).$$

Choose the transversal section

$$s(a, d, f) = (a, 0, 0, d, 0, f).$$

Then

$$s(a, d, f)s(a', d', f')$$

equals

$$(a + a', ad', 0, d + d', df', f + f').$$

But

$$s(a + a', d + d', f + f') = (a + a', 0, 0, d + d', 0, f + f').$$

Therefore

$$s(x)s(y) \neq s(x + y).$$

The discrepancy is exactly

$$(ad', df').$$

Thus the chosen section fails to be a homomorphism.

The extension therefore does not behave like a naive semidirect decomposition at the level of these coordinates.

5 Fourier Phase Geometry

Now Fourier transform the (b, e) -coordinates.

Let

$$\beta, \epsilon \in \mathbb{Z}_5$$

be the corresponding Fourier dual variables.

Then the cocycle translations become phases:

$$b \mapsto b + ad' \implies \omega_5^{\beta ad'},$$

and

$$e \mapsto e + df' \implies \omega_5^{\epsilon df'}.$$

Similarly, the correction

$$c \mapsto c + ae' + bf'$$

becomes

$$\omega_5^{\gamma(ae' + bf')},$$

where γ is the Fourier dual variable to c .

Thus the splitting obstruction becomes a Fourier phase interaction.

The nonabelian correction terms are reorganized into computable phase couplings between Fourier layers.

This yields the schematic principle:

$$\boxed{\text{splitting obstruction} \longrightarrow \text{cocycle correction} \longrightarrow \text{Fourier phase geometry.}}$$

6 Periodic Structure of Fourier Tower Operators

One natural question is whether Fourier Tower operators satisfy finite-order relations analogous to the classical identity

$$F_N^4 = I$$

for ordinary finite Fourier transforms.

At the local Fourier level, one always has periodicity, while the global Fourier Tower operator depends additionally on cocycle phase interactions.

6.1 Local Fourier Periodicity

Let Q be a finite abelian group and let

$$F_Q : \mathbb{C}[Q] \rightarrow \mathbb{C}[Q]$$

denote the ordinary finite Fourier transform.

For finite abelian groups one has the classical identity

$$F_Q^2 f(x) = f(-x),$$

and therefore

$$F_Q^4 = I.$$

Now consider a solvable or nilpotent tower

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

with abelian quotient layers

$$Q_i = G_i / G_{i-1}.$$

The corresponding local Fourier transform is

$$F_{\text{loc}} = \bigotimes_i F_{Q_i}.$$

Since each factor satisfies

$$F_{Q_i}^4 = I,$$

it follows that

$$\boxed{F_{\text{loc}}^4 = I.}$$

Thus the purely abelian part of the Fourier Tower always has period 4.

6.2 Cocycle-Corrected Fourier Towers

The full Fourier Tower transform is not simply the local Fourier operator.

Indeed, the nonabelian extension structure introduces cocycle corrections coupling the quotient layers.

Schematically, the Fourier Tower operator has the form

$$F_{\mathcal{T}} = C_{\omega} F_{\text{loc}},$$

where:

- F_{loc} is the tensor product of local abelian Fourier transforms,
- C_{ω} is a cocycle-controlled phase operator.

The operator C_{ω} acts diagonally on Fourier character sectors through phase factors induced by cocycle corrections.

For example, if

$$s(q)s(q') = \omega(q, q')s(qq'),$$

then after Fourier transformation of the lower layer, the cocycle contributes the phase

$$\chi(\omega(q, q')).$$

Thus the full Fourier Tower dynamics combines:

Fourier duality + cocycle phase propagation.

Proposition 1 (Periodic Structure in Finite Fourier Towers). *Let*

$$F_{\mathcal{T}} = C_{\omega} F_{\text{loc}}$$

be a Fourier Tower operator associated with a finite solvable tower.

Assume:

- *the local Fourier transform satisfies*

$$F_{\text{loc}}^4 = I,$$

- *the cocycle phase operator C_{ω} has finite order,*
- *and C_{ω} commutes with F_{loc} .*

Then $F_{\mathcal{T}}$ has finite order. More precisely, there exists an integer $m \geq 1$ such that

$$F_{\mathcal{T}}^m = I.$$

In particular, one may choose m to be a common multiple of

$$4 \quad \text{and} \quad \text{ord}(C_{\omega}).$$

Proof. Since

$$F_{\text{loc}}^4 = I,$$

the operator F_{loc} has finite order dividing 4.

By assumption, there exists an integer $r \geq 1$ such that

$$C_\omega^r = I.$$

Since C_ω commutes with F_{loc} , one has

$$F_{\mathcal{T}}^m = (C_\omega F_{\text{loc}})^m = C_\omega^m F_{\text{loc}}^m.$$

Choose m to be a common multiple of 4 and r , for example

$$m = \text{lcm}(4, r).$$

Then

$$C_\omega^m = I, \quad F_{\text{loc}}^m = I.$$

Therefore

$$F_{\mathcal{T}}^m = C_\omega^m F_{\text{loc}}^m = I.$$

Hence $F_{\mathcal{T}}$ has finite order. □

Remark 1. *The commutativity assumption between*

$$C_\omega \quad \text{and} \quad F_{\text{loc}}$$

is highly nontrivial.

In general, cocycle phase operators need not commute with the local Fourier transform.

Consequently, even though

$$F_{\text{loc}}^4 = I,$$

one may have

$$F_{\mathcal{T}}^4 \neq I.$$

Thus the full Fourier Tower dynamics is generally more complicated than ordinary Fourier inversion.

The noncommuting case is expected to produce richer cocycle-Fourier dynamics.

This suggests that the interaction between Fourier duality and cocycle propagation may itself encode structural information about the underlying nonabelian tower.

6.3 Application to $UT_4(\mathbb{Z}_5)$

For the nilpotent group

$$UT_4(\mathbb{Z}_5),$$

the cocycle corrections are

$$ad', \quad df', \quad ae' + bf'.$$

After Fourier transformation of the corresponding layers, these produce phase factors

$$\omega_5^{\beta ad'}, \quad \omega_5^{\epsilon df'}, \quad \omega_5^{\gamma(ae' + bf')},$$

where

$$\omega_5 = e^{2\pi i/5}.$$

The cocycle phase factors are built from fifth roots of unity. Consequently, the cocycle phase operator has finite order.

Since the local Fourier transform satisfies

$$F_{\text{loc}}^4 = I,$$

the full Fourier Tower operator combines

order-4 Fourier duality

with

finite-order cocycle phase dynamics.

The precise order of the full Fourier Tower operator depends on the commutation relations between the local Fourier part and the cocycle phase part.

This illustrates the general principle:

nonabelianity = interaction between Fourier layers and cocycle phases.

7 Internal Splitting, Clifford Splitting, and Symplectic Structures

It is important to distinguish two different notions of splitting appearing in this work.

7.1 Internal versus External Splitting

The first concerns internal group extensions of the form

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1,$$

where failure of multiplicativity of transversal sections produces cocycle corrections inside the group itself.

This is the primary object studied by the Fourier Tower framework.

The second concerns external automorphism extensions such as

$$1 \rightarrow H(N) \rightarrow \text{Cliff}(N) \rightarrow Sp(2n, \mathbb{Z}_N) \rightarrow 1,$$

where one studies whether symplectic automorphisms admit compatible lifts to Clifford operators.

These two splitting problems are conceptually different.

Fourier Towers primarily analyze the first phenomenon: the recursive internal cocycle structure of solvable and nilpotent groups.

However, the appearance of cocycle phase corrections suggests possible analogies with non-splitting phenomena in Clifford extensions.

7.2 Relation with Symplectic and Clifford Structures

The Fourier Tower viewpoint should be distinguished from the classical symplectic and Clifford constructions associated with the Heisenberg group.

Consider the finite Heisenberg group

$$H_p.$$

It fits into a central extension

$$1 \rightarrow Z(H_p) \rightarrow H_p \rightarrow V \rightarrow 1,$$

where

$$V \cong \mathbb{F}_p^{2n}.$$

This extension is controlled by the Heisenberg cocycle

$$\omega(v, v').$$

The extension itself is nontrivial: the cocycle measures the failure of multiplicative splitting of the Heisenberg group into independent abelian layers.

Thus the Heisenberg structure is intrinsically cocyclic.

However, the symplectic group

$$Sp(2n, \mathbb{F}_p)$$

acts by automorphisms on the Heisenberg group while preserving the symplectic cocycle structure.

This produces the semidirect product

$$H_p \rtimes Sp(2n, \mathbb{F}_p),$$

which forms the algebraic origin of the finite Clifford group.

Therefore two distinct structures coexist:

- an internal cocycle structure coming from the non-splitting Heisenberg extension,
- and an external symplectic action acting by automorphisms.

The Fourier Tower framework focuses primarily on the first phenomenon.

Indeed, the Fourier Tower does not study automorphism groups acting externally on the Heisenberg structure. Instead, it analyzes how the internal cocycle corrections themselves generate interactions between Fourier layers.

From this viewpoint:

Heisenberg cocycle $\longrightarrow$ Fourier phase interaction.

The Clifford or symplectic picture acts globally on the Heisenberg structure, whereas the Fourier Tower framework resolves the internal nonabelianity recursively through local Fourier decompositions and cocycle phase couplings.

7.3 Failure of Clifford Splitting

For finite-dimensional quantum systems, the Clifford group is often described schematically as a semidirect extension involving the Heisenberg group and a symplectic group:

$$\text{Cliff}(N) \sim H(N) \rtimes SL(2, \mathbb{Z}_N).$$

However, this splitting is subtle and may fail globally.

In particular, in some finite-dimensional settings, arithmetic conditions such as divisibility by 4 can be related to splitting obstructions.

In such cases, one encounters cocycle obstructions preventing a global splitting of the Clifford extension.

Thus the lifting of symplectic transformations to Clifford operators is only projective in general.

This phenomenon is conceptually related to the Fourier Tower viewpoint.

Indeed, both situations involve:

- a lower algebraic structure,
- a failure of exact multiplicative lifting,
- cocycle corrections,
- and resulting phase ambiguities.

The difference is that in Clifford theory the cocycle obstruction appears at the automorphism level, while in Fourier Towers it appears internally through recursive quotient decompositions.

This suggests a broader principle:

$$\text{cocycle obstructions} = \text{phase corrections associated with lifting problems.}$$

From this perspective, Fourier Towers may provide a recursive harmonic framework for understanding how non-splitting phenomena propagate through layered algebraic structures.

8 Structural Significance of the Fourier Tower Viewpoint

The purpose of the Fourier Tower framework is not merely to rewrite nilpotent multiplication laws in Fourier variables.

Rather, the framework attempts to reinterpret cocycle obstructions and non-splitting phenomena as structured interactions between local Fourier layers.

Several conceptual features emerge from this viewpoint.

8.1 Internal versus External Nonabelianity

The Fourier Tower framework distinguishes two different sources of nonabelian structure.

The first is internal nonabelianity, arising from cocycle corrections inside solvable or nilpotent extensions

$$1 \rightarrow N \rightarrow G \rightarrow Q \rightarrow 1.$$

The second is external symmetry nonabelianity, arising from automorphism groups acting on a lower structure, such as

$$H(N) \rtimes Sp(2n, \mathbb{Z}_N).$$

The Fourier Tower framework focuses primarily on the first phenomenon.

From this viewpoint, the essential structure is not the external automorphism action itself, but rather the recursive cocycle corrections generated inside the extension tower.

8.2 Cocycles as Fourier Dynamical Objects

In classical extension theory, cocycles measure the failure of multiplicative splitting.

In the Fourier Tower framework, cocycles acquire an additional harmonic interpretation.

Indeed, after Fourier transformation of an abelian layer, a cocycle

$$\omega(q, q')$$

acts through the phase

$$\chi(\omega(q, q')).$$

Thus cocycles become dynamical phase interactions between local Fourier sectors.

Schematically,

$$\boxed{\text{extension cocycle} \longrightarrow \text{Fourier phase dynamics.}}$$

This interpretation shifts cocycles from purely algebraic correction terms to harmonic coupling mechanisms.

8.3 The Role of $UT_4(\mathbb{Z}_5)$

The importance of

$$UT_4(\mathbb{Z}_5)$$

is not merely that it is larger than the Heisenberg group.

Its significance is that several cocycle obstructions coexist simultaneously across different nilpotent layers.

The multiplication law contains:

$$ad', \quad df', \quad ae' + bf',$$

which generate interacting phase corrections after Fourier transformation.

Thus

$$UT_4(\mathbb{Z}_5)$$

may be viewed as an iterated cocycle obstruction system.

This makes it a natural laboratory for studying recursive Fourier-cocycle interactions.

8.4 Periodic Fourier-Cocycle Dynamics

The Fourier Tower viewpoint also introduces a dynamical aspect.

The local Fourier transforms satisfy the classical periodicity relation

$$F_{\text{loc}}^4 = I.$$

However, the full Fourier Tower operator

$$F_{\mathcal{T}} = C_{\omega} F_{\text{loc}}$$

contains additional cocycle phase interactions.

Therefore the global dynamics is no longer purely Fourier inversion, but rather a coupled cocycle-Fourier evolution.

This suggests that Fourier Towers may also be viewed as finite dynamical systems generated by recursive cocycle phase propagation, whenever the corresponding operators satisfy suitable finite-order conditions.

8.5 Relation with Clifford-Type Obstructions

The comparison with Clifford extensions suggests a broader analogy.

In Clifford theory, one encounters cocycle obstructions preventing global lifting of symplectic automorphisms.

In Fourier Towers, one encounters cocycle obstructions preventing global multiplicative splitting of extension towers.

In both situations:

- lifting fails globally,
- cocycle corrections appear,
- and phase ambiguities are generated.

This suggests the broader heuristic principle:

$$\boxed{\text{cocycle obstructions} = \text{phase corrections associated with lifting problems.}}$$

The Fourier Tower framework therefore suggests a recursive harmonic interpretation of non-splitting phenomena in solvable and nilpotent structures.

9 Possible Connections with Quantum Circuit Structures

The Fourier Tower framework also suggests possible connections with finite-dimensional quantum circuit architectures.

At a structural level, quantum circuits are built from:

- local Fourier transforms,
- phase gates,
- controlled interactions,
- and noncommuting operator layers.

These same ingredients naturally appear in the Fourier Tower framework.

Indeed, the local Fourier operators

$$F_{Q_i}$$

associated with abelian quotient layers behave analogously to local quantum Fourier transform blocks, while the cocycle phase operators

$$C_\omega$$

act through phase corrections coupling different Fourier sectors.

Schematically, the Fourier Tower operator

$$F_{\mathcal{T}} = C_\omega F_{\text{loc}}$$

resembles a recursive circuit decomposition consisting of:

- local Fourier transforms,
- followed by cocycle-controlled phase interactions.

From this viewpoint, cocycle obstructions may be interpreted as phase-propagation mechanisms between different layers of a recursive nonabelian structure.

9.1 Recursive Phase Couplings

The nilpotent example

$$UT_4(\mathbb{Z}_5)$$

is particularly interesting because several cocycle corrections coexist simultaneously:

$$ad', \quad df', \quad ae' + bf'.$$

After Fourier transformation, these become interacting phase terms:

$$\omega_5^{\beta ad'}, \quad \omega_5^{\epsilon df'}, \quad \omega_5^{\gamma(ae' + bf')}.$$

Thus the group generates a hierarchy of coupled phase interactions rather than a single global Fourier phase.

This suggests an interpretation of Fourier Towers as recursive phase-coupling systems in which:

- local Fourier transforms act on quotient layers,
- cocycles generate controlled phase propagation,
- and higher nilpotent corrections produce interactions between multiple Fourier sectors.

9.2 Relation with Clifford-Type Structures

The classical Clifford framework is closely related to the Heisenberg group together with symplectic automorphisms.

In that setting, one essentially studies a single cocycle structure associated with the Heisenberg extension.

The Fourier Tower framework suggests a possible higher-order analogue: instead of a single cocycle phase layer, one obtains recursive cocycle interactions distributed across several nilpotent layers.

Schematically:

$$\boxed{\text{Heisenberg-Clifford structure} \approx \text{one primary Heisenberg cocycle layer} + \text{symplectic action},}$$

whereas

$$\boxed{\text{Fourier Tower structure} = \text{recursive cocycle phase hierarchy}.}$$

This viewpoint may eventually provide a useful language for describing hierarchical phase propagation in finite-dimensional quantum systems.

9.3 Conceptual Perspective

The present work does not propose a new quantum algorithm or a complete quantum computational model.

Rather, the goal is conceptual: to reinterpret nonabelian extension data as recursive Fourier-phase interactions.

From this perspective, Fourier Towers may provide:

- a harmonic description of cocycle propagation,
- a recursive organization of nilpotent phase interactions,
- and a possible bridge between extension theory, Fourier analysis, and finite quantum phase geometry.

10 Toward an $F + \Phi + H$ Decomposition

The example of $UT_4(\mathbb{Z}_5)$ suggests a decomposition principle which is not simply inherited from the usual Heisenberg–Clifford framework.

In the Clifford–Heisenberg setting, the basic structure is governed by a finite Heisenberg group together with an external symplectic action.

This leads to a decomposition based on:

Heisenberg phase space + symplectic propagation.

By contrast, $UT_4(\mathbb{Z}_5)$ contains several internal nilpotent layers and several nested cocycle corrections.

This suggests a Fourier Tower decomposition of the form

$$\boxed{UT_4 \rightsquigarrow F + \Phi + H,}$$

where:

F = local Fourier layers,

Φ = cocycle phase interactions,

and

H = higher nilpotent propagation.

Here F corresponds to the local Fourier transforms on the abelian quotient layers

$$\mathbb{Z}_5^3, \quad \mathbb{Z}_5^2, \quad \mathbb{Z}_5.$$

The term Φ records the cocycle phase interactions arising from the correction terms

$$ad', \quad df', \quad ae' + bf'.$$

After Fourier transformation, these become phase factors

$$\omega_5^{\beta ad'}, \quad \omega_5^{\epsilon df'}, \quad \omega_5^{\gamma(ae' + bf')}.$$

The term H represents the higher nilpotent propagation between successive layers of the lower central tower.

Thus the nonabelian structure is not organized by a single symplectic phase space, as in the Heisenberg–Clifford case, but by a recursive hierarchy of Fourier layers and cocycle phases.

In this sense, the $F + \Phi + H$ decomposition may be viewed as a Fourier Tower analogue of Clifford–Heisenberg decompositions, but one adapted to higher nilpotent symmetry rather than to a single Heisenberg central extension.

Schematically:

$$\boxed{\text{Clifford–Heisenberg} = \text{Fourier} + \text{single cocycle phase} + \text{symplectic action},}$$

whereas

$$\boxed{UT_4 \text{ Fourier Tower} = F + \Phi + H.}$$

This suggests that $UT_4(\mathbb{Z}_5)$ may provide a prototype for Fourier decompositions beyond the standard Heisenberg–Clifford setting, based on higher nilpotent cocycle geometry.

Proposition 2 ($F + \Phi + H$ Decomposition of Fourier Tower Data for $UT_4(\mathbb{Z}_5)$). *For $UT_4(\mathbb{Z}_5)$, the Fourier Tower data decomposes into three components*

$$F, \quad \Phi, \quad H,$$

which are summarized conceptually by

$$F + \Phi + H.$$

Here:

$$F = F_{\mathbb{Z}_5^3} \otimes F_{\mathbb{Z}_5^2} \otimes F_{\mathbb{Z}_5},$$

$$\Phi = \omega_5^{\beta ad'} \omega_5^{\epsilon df'} \omega_5^{\gamma(ae' + bf')},$$

and H is the higher nilpotent propagation encoded by the lower central tower

$$UT_4(\mathbb{Z}_5) = \Gamma_1 \triangleright \Gamma_2 \triangleright \Gamma_3 \triangleright \{e\}.$$

Thus the nonabelian multiplication data decomposes into:

$local \text{ Fourier layers} + cocycle \text{ phase interactions} + higher \text{ nilpotent propagation}.$

Proof. The lower central series of $UT_4(\mathbb{Z}_5)$ has quotient layers

$$\Gamma_1/\Gamma_2 \cong \mathbb{Z}_5^3,$$

$$\Gamma_2/\Gamma_3 \cong \mathbb{Z}_5^2,$$

and

$$\Gamma_3 \cong \mathbb{Z}_5.$$

Therefore the local abelian Fourier part is

$$F = F_{\mathbb{Z}_5^3} \otimes F_{\mathbb{Z}_5^2} \otimes F_{\mathbb{Z}_5}.$$

This gives the F -component.

Now write an element of $UT_4(\mathbb{Z}_5)$ as

$$(a, b, c, d, e, f),$$

with multiplication

$$(a, b, c, d, e, f)(a', b', c', d', e', f')$$

$$= (a + a', b + b' + ad', c + c' + ae' + bf', d + d', e + e' + df', f + f').$$

The nonabelian correction terms are precisely

$$ad', \quad df', \quad ae' + bf'.$$

They translate the higher coordinates:

$$b \mapsto b + ad',$$

$$e \mapsto e + df',$$

and

$$c \mapsto c + ae' + bf'.$$

After Fourier transformation in the b, e, c -coordinates, translations become phase multiplications.

If β, ϵ, γ are the Fourier dual variables of b, e, c , then the corrections become

$$\omega_5^{\beta ad'},$$

$$\omega_5^{\epsilon df'},$$

and

$$\omega_5^{\gamma(ae'+bf')}.$$

Hence the total cocycle phase contribution is

$$\Phi = \omega_5^{\beta ad'} \omega_5^{\epsilon df'} \omega_5^{\gamma(ae'+bf')}.$$

This gives the Φ -component.

Finally, the correction terms are not independent. They are organized by the nilpotent tower:

$$\Gamma_1/\Gamma_2 \longrightarrow \Gamma_2/\Gamma_3 \longrightarrow \Gamma_3.$$

The first-order layer (a, d, f) produces second-order corrections (b, e) , and the second-order layer contributes to the central coordinate c .

This propagation through the lower central tower is the H -component.

Therefore the Fourier Tower data associated with $UT_4(\mathbb{Z}_5)$ decomposes into the three components

$$F, \quad \Phi, \quad H,$$

summarized conceptually as

$$F + \Phi + H,$$

where F is the local Fourier part, Φ is the cocycle phase part, and H is the higher nilpotent propagation. □

Remark 2. *This proposition should not be interpreted as a group isomorphism*

$$UT_4(\mathbb{Z}_5) \cong F \times \Phi \times H.$$

Rather, it is a decomposition of the Fourier Tower data associated with the multiplication law of $UT_4(\mathbb{Z}_5)$.

It shows that the nonabelian structure separates into local Fourier layers, cocycle phase corrections, and higher nilpotent propagation.

11 Conclusion

We developed a Fourier interpretation of splitting obstructions and cocycle corrections in solvable and nilpotent group towers.

The main principle is that failure of multiplicative splitting naturally generates cocycle corrections, and Fourier transformation converts these corrections into phase interactions between character layers.

The example

$$UT_4(\mathbb{Z}_5)$$

demonstrates explicitly how nested cocycle corrections produce hierarchical Fourier phase couplings.

The importance of $UT_4(\mathbb{Z}_5)$ is not merely that it is a larger nilpotent group than the Heisenberg group.

Its significance is that several cocycle obstructions coexist simultaneously and interact across different nilpotent layers.

Thus $UT_4(\mathbb{Z}_5)$ provides a natural laboratory for studying recursive cocycle Fourier interactions.

The resulting framework suggests the conceptual principle

$$\boxed{\text{nonabelianity} = \text{recursive interaction between local Fourier layers.}}$$

More broadly, Fourier Towers may provide a recursive alternative to global semidirect-product decompositions and a useful language for organizing nilpotent harmonic analysis, cocycle geometry, and layered quantum symmetries.

The proposed $F + \Phi + H$ viewpoint summarizes this idea:

$$F = \text{Fourier layers}, \quad \Phi = \text{phase cocycles}, \quad H = \text{higher nilpotent propagation.}$$

This provides a conceptual bridge between extension theory, Fourier analysis, finite phase geometry, and possible quantum circuit structures.

References

- [1] J.-P. Serre, *Linear Representations of Finite Groups*, Springer, 1977.
- [2] A. Terras, *Fourier Analysis on Finite Groups and Applications*, Cambridge University Press, 1999.
- [3] G. W. Mackey, *Induced Representations of Groups and Quantum Mechanics*, W. A. Benjamin, 1968.
- [4] A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.
- [5] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.
- [6] D. Appleby, “Symmetric Informationally Complete Measurements of Arbitrary Rank,” arXiv:quant-ph/0611260.
- [7] D. Gross, “Hudson’s theorem for finite-dimensional quantum systems,” *Journal of Mathematical Physics*, 47, 122107 (2006).